

Student 14 – Vertex Cover Problem

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Q14. Vertex Cover Problem (NP-Complete / Constraint Satisfaction Problem)

Answer

The **Vertex Cover Problem** is the problem of finding a minimum set of vertices in a graph such that every edge has at least one endpoint selected.

1. Constraints

Let the graph be $G=(V,E)$. For every vertex v , define a binary variable:

- $x_v = 1 \rightarrow$ vertex v is selected.
- $x_v = 0 \rightarrow$ vertex v is not selected.

For every edge $(u,v) \in E$, the edge coverage constraint is:

$$x_u + x_v \geq 1$$

This means at least one endpoint of every edge must be included in the vertex cover. The objective is to minimize $\sum x_v$.

2. Backtracking Approach

Backtracking explores different subsets of vertices and checks whether they form a valid vertex cover.

1. Start with an empty set of selected vertices.
2. Select a vertex or exclude it.
3. Check whether all edges are covered.
4. If a valid cover is found, compare its size with the current minimum.
5. If the current selection is already larger than the best solution, prune that branch.
6. Continue until all possible useful combinations are explored.

3. Simple Pseudocode

```
VertexCover(V, E, selected, index):  
  
    if all edges are covered:  
        update minimum solution  
        return  
  
    if size(selected) >= minimum:  
        return // Pruning  
  
    choose an uncovered edge (u, v)  
  
    include u  
    VertexCover(V, E, selected + {u}, index + 1)  
  
    exclude u and include v  
    VertexCover(V, E, selected + {v}, index + 1)
```

4. Example

Consider the graph:



C ----- D

Edges are: (A,B), (A,C), (B,D), (C,D).

A minimum vertex cover can be: **{A,D}**.

Check:

- A-B → covered by A
- A-C → covered by A
- B-D → covered by D
- C-D → covered by D

Therefore, {A,D} is a valid minimum vertex cover of size 2.

5. Pruning

Pruning avoids unnecessary combinations. If the current selected set already has the same or more vertices than the best solution found, that branch cannot produce a better answer, so it is stopped.

6. Why is Vertex Cover NP-Complete?

The **decision version** asks: “Does the graph have a vertex cover of size at most k?” It is in NP because a proposed vertex cover can be verified in polynomial time by checking every edge. It is NP-hard because known NP-complete problems can be reduced to it. Therefore, the decision version of Vertex Cover is NP-Complete.

Conclusion

The backtracking method systematically explores possible vertex subsets while using edge-coverage constraints and pruning to reduce unnecessary searches. The final selected set covers every edge and, when its size is minimum, it is the required minimum vertex cover.